

Robot Learning Engineer Master Curriculum

Day 2 Packet - July 6, 2026

Target Role: Robot Learning / Embodied AI Engineer

Today connects beginner Python to robotics-style decision logic. You will practice conditionals through robot battery, CTWD, temperature, and weld-quality decisions, while continuing interview-oriented DSA and GitHub documentation.

Day Overview

Block	Time	Focus	Main Output
0	As needed	Cisco portal first	Apply before deadline if still pending
1	2 hours	Python conditionals	5 robotics-flavored decision programs
2	1.5 hours	DSA arrays/hash maps	Two Sum and Contains Duplicate explained
3	45 minutes	GitHub portfolio system	Clean Day02 commit pushed
4	1 hour	Robot learning direction	Notes on RL, imitation learning, simulation, embodied AI
5	10 minutes	Journal	Day02 report written

Primary Resources for Today

Use only the resources below today. Do not jump between random tutorials. Resource discipline is part of the training.

Track	Resource	Use Today
Python course	CS50P Week 1: Conditionals	Watch/read Conditionals. Focus on if, elif, else, comparison operators, and boolean logic.
Python documentation	Python control flow tools	Read only the if-statement part today and skim indentation examples.
DSA practice	LeetCode Two Sum	Write brute force and optimized hash map idea.
DSA practice	LeetCode Contains Duplicate	Practice using a set/hash table to detect repeats.
Git book	Pro Git official online book	Read or review Section 2.1: Getting a Git Repository.
Robot learning overview	Berkeley CS285 Deep RL course	Only inspect the syllabus/topics. Do not start deep RL yet.

Book note: Use legal access only. If a paid book is unavailable, use the official course/documentation and complete the code deliverables.

Block 0 - Submission Priority

Goal: Do not lose opportunities because of study scheduling.

If any internship application deadline is active before noon, submit the application before studying. After submission, return to this plan.

For future weeks, application deadlines override normal study order. Career execution is part of the roadmap.

Block 1 - Python Conditionals for Robotics Decisions

Goal: Use if/elif/else to make clean engineering decisions.

- Watch/read CS50P Week 1: Conditionals.
- Learn comparison operators: ==, !=, <, <=, >, >=.
- Learn boolean logic: and, or, not.
- Write each program yourself. Do not use AI-generated code as final code.

File	Program Requirement
grade_checker.py	Input marks and print grade. Add invalid input handling if possible.
temperature_status.py	Input temperature and print cold, normal, hot, or danger.
robot_battery_status.py	Input battery percentage and print Critical, Low, Normal, or Full.
ctwd_status_checker.py	Input CTWD value and print Too close, Ideal, or Too far.
weld_quality_decision.py	Input travel speed, work angle, and CTWD. Print Good, Average, or Bad.

```
robot-learning-journey/  
  Day02/  
    grade_checker.py  
    temperature_status.py  
    robot_battery_status.py  
    ctwd_status_checker.py  
    weld_quality_decision.py  
    dsa_notes.md  
    robot_learning_notes.md  
    journal.md
```

Block 2 - DSA Arrays and Hash Maps

Goal: Understand the difference between brute force and optimized lookup.

For each problem write: brute-force idea, optimized idea, time complexity, space complexity, and one edge case.

Problem	Required Notes
Two Sum	Brute force $O(n^2)$, hash map $O(n)$, why hash maps help.
Contains Duplicate	Sorting idea vs set idea, time complexity, space complexity.

Block 3 - GitHub Portfolio System

Goal: Make your GitHub show daily evidence of work.

```
git status
git add .
git commit -m "Day 2: Python conditionals and DSA basics"
git log --oneline
git push
```

Block 4 - Robot Learning Direction Notes

Goal: Understand the map before going deep.

Open Berkeley CS285 and only inspect the topics. Write simple answers in robot_learning_notes.md. No equations today.

- What is reinforcement learning?
- Why is RL useful for robots?
- What is imitation learning?
- Why do robots need simulation?
- What is the difference between normal AI and embodied AI?

Completion Checklist

[] CS50P Week 1 Conditionals completed or carefully read.
[] 5 Python conditional programs completed in Day02 folder.
[] Two Sum solved or seriously attempted with notes.
[] Contains Duplicate solved or seriously attempted with notes.
[] dsa_notes.md written with complexity notes.
[] robot_learning_notes.md written in simple language.
[] Day02 Git commit created and pushed.
[] journal.md completed and progress pasted into chat.

What To Send Me After Finishing

Copy-paste this report into chat. This is how we adjust the next day without losing context.

Day 2 completed report:
1. Hours studied:
2. Python files completed:
3. DSA problems completed:
4. GitHub link/commit:
5. What was easy:
6. What was hard:
7. What I skipped:
8. Questions:

Quality Rules

- Do not copy code without understanding it yourself.
- Do not hide skipped tasks. Honest reporting makes the roadmap accurate.
- Every study day ends with code, notes, a GitHub commit, and a journal entry.
- Certificates are secondary. GitHub evidence, projects, research-quality work, and interview skill matter more.

Source Notes and Resource Policy

Day 2 uses conditionals because robotics systems constantly make threshold decisions: battery health, sensor range, CTWD status, safety state, and action selection. This is basic Python, but the examples are chosen from your robotics/AR-VR welding direction.